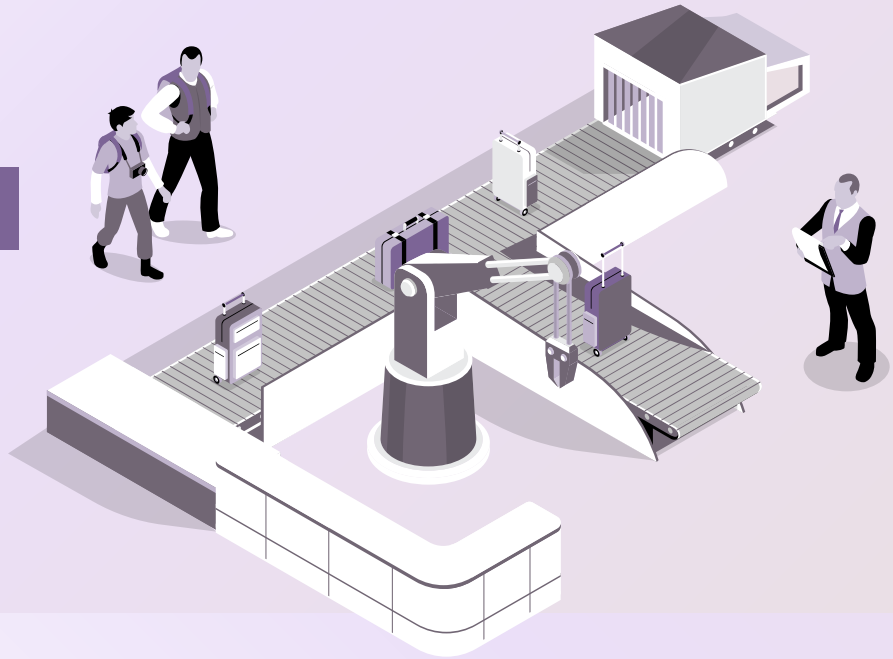
AIRPORT NETWORK

Network Design Case Study

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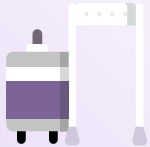


REQUIREMENTS VS DESIGN



REQUIREMENTS

Secure department separation, shared services, Wi-Fi and managed IP addressing.



OUR DESIGN

9 VLANs, router-on-a-stick, DHCP on R1, OSPF to R2-Edge, servers in VLAN 90, WPA2 Wi-Fi.

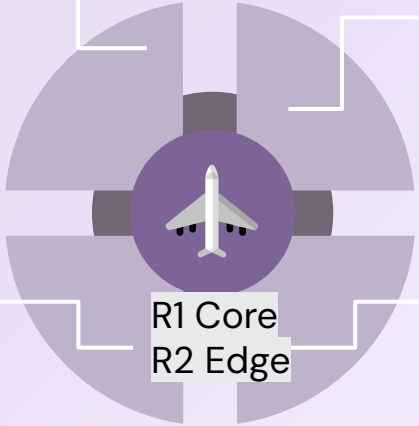
Network Topology by Department

ADMINISTRATION

Admin and check-in PCs connected via SW1-Admin

OPERATIONS & SECURITY

Security and baggage PCs connected via SW2-Operations



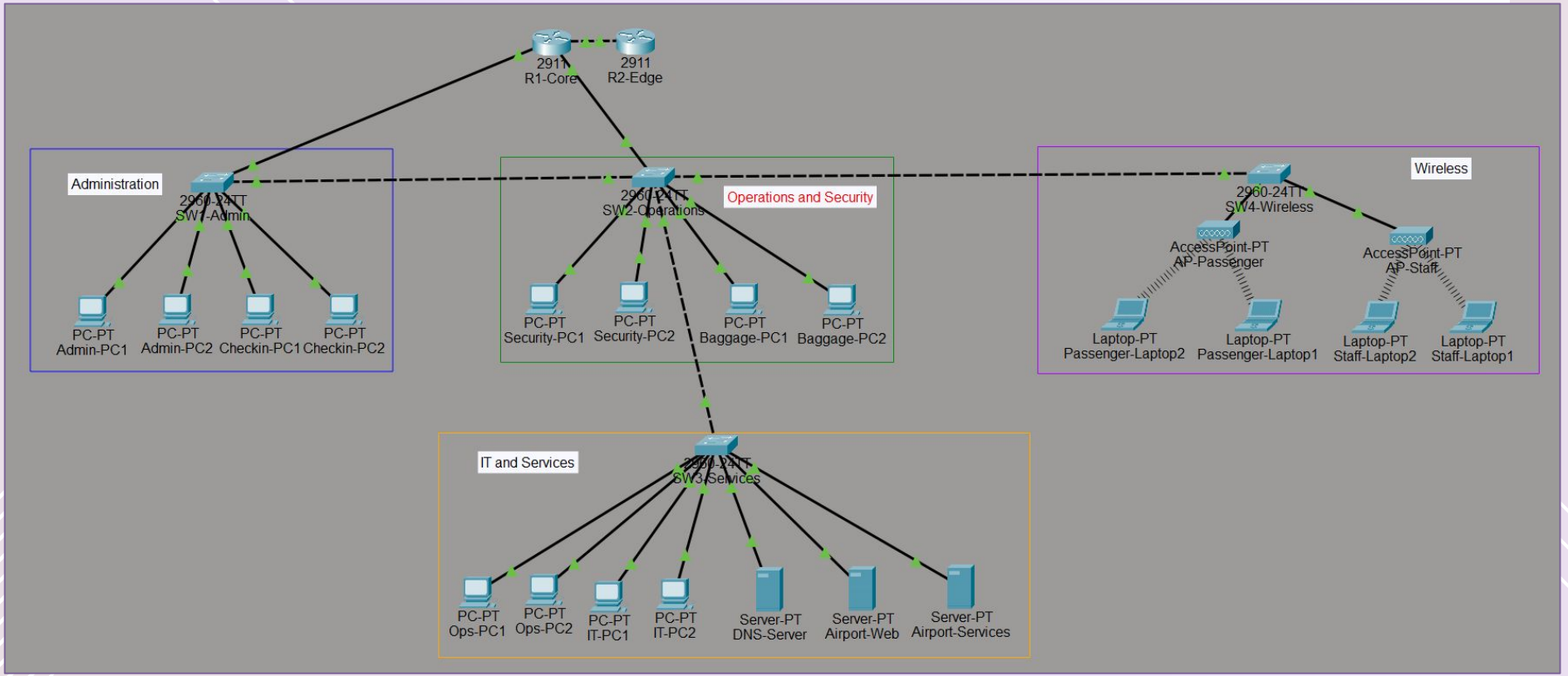
IT & SERVICES

IT/Ops PCs, DNS, web and airport servers via SW3-Services

WIRELESS

Passenger and staff Wi-Fi via SW4-Wireless and 2 APs

NETWORK TOPOLOGY



DESIGN APPROACH

VLANS

9 VLANs: Admin, Check-in, Security, Baggage, Ops, IT, Wi-Fi x2, Servers

INTER-VLAN

R1-Core 802.1Q subinterfaces (router-on-a-stick)

ADDRESSING

One /24 per VLAN. DHCP for clients, static IPs for servers

ROUTING

OSPF between R1-Core and R2-Edge over 10.0.0.0/30

SERVICES

DNS, Web, FTP and Email hosted in server VLAN 90

WIRELESS

Staff (VLAN 70) and Passenger (VLAN 80) SSIDs on 2 APs

VLAN AND IP ADDRESSING

VLAN	Department	Subnet	Gateway (R1-Core)	Access switch
10	Administration	192.168.10.0/24	192.168.10.1	SW1-Admin
20	Check-in & Ticketing	192.168.20.0/24	192.168.20.1	SW1-Admin
30	Security	192.168.30.0/24	192.168.30.1	SW2-Operations
40	Baggage Handling	192.168.40.0/24	192.168.40.1	SW2-Operations
50	Airport Operations	192.168.50.0/24	192.168.50.1	SW3-Services
60	IT Department	192.168.60.0/24	192.168.60.1	SW3-Services
70	Staff Wi-Fi	192.168.70.0/24	192.168.70.1	SW4-Wireless
80	Passenger Wi-Fi	192.168.80.0/24	192.168.80.1	SW4-Wireless
90	Servers	192.168.90.0/24	192.168.90.1	SW3-Services

- **Router link:** R1 Gi0/0 10.0.0.1/30 – R2 Gi0/0 10.0.0.2/30
- **Servers (static):** DNS 192.168.90.10 · Web 192.168.90.20 · Airport-Services 192.168.90.30
- **DHCP:** first 10 addresses of each client subnet excluded; DNS option 192.168.90.10

SWITCHING: VLANS AND TRUNKS

- **Local VLAN configuration:** each switch defines the VLANs its own hosts need: SW1 (10, 20), SW2 (30–90), SW3 (50, 60, 90), SW4 (70, 80)
- **Local access ports:** hosts attach as untagged access ports, e.g. SW1 Fa0/3–4 in VLAN 10, SW2 Fa0/5–6 in VLAN 30 and Fa0/7–8 in VLAN 40, SW3 Fa0/6–8 in VLAN 90
- **Trunk transport:** SW2 Fa0/1–4 run IEEE 802.1Q and carry tagged frames for the active VLANs between devices
- **Result:** VLAN membership is set per switch, trunks carry it across the network, and R1-Core routes between VLANs

SW2 trunk ports

```
SW2-Operations(config)#do show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1
Fa0/4       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Fa0/2       1-1005
Fa0/3       1-1005
Fa0/4       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,30,40,50,60,70,80,90
Fa0/2       1,30,40,50,60,70,80,90
Fa0/3       1,30,40,50,60,70,80,90
Fa0/4       1,30,40,50,60,70,80,90

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,30,40,50,60,70,80,90
Fa0/2       none
Fa0/3       none
 --More--
```

SW3 VLAN assignment

```
SW3-Services(config)#do show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
50   OPERATIONS                       active    Fa0/2, Fa0/3
60   IT                               active    Fa0/4, Fa0/5
90   SERVERS                          active    Fa0/6, Fa0/7, Fa0/8
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW3-Services(config)#
```

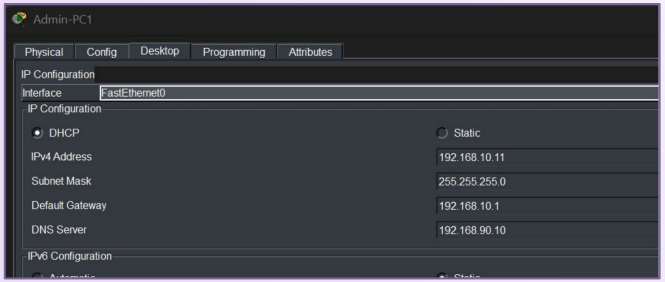
INTER-VLAN ROUTING AND DHCP

R1-Core subinterfaces (gateways)

```
Router(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.1        YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  up                    up
GigabitEthernet0/1.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/1.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  up                    up
GigabitEthernet0/2.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/2.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/2.50  192.168.50.1    YES manual up                    up
GigabitEthernet0/2.60  192.168.60.1    YES manual up                    up
GigabitEthernet0/2.70  192.168.70.1    YES manual up                    up
GigabitEthernet0/2.80  192.168.80.1    YES manual up                    up
GigabitEthernet0/2.90  192.168.90.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router(config)#
```

- **Router-on-a-stick:** Gi0/1.10 and .20 serve VLANs 10 and 20; Gi0/2.30–.90 serve VLANs 30–90. Each subinterface is the .1 gateway of its /24
- **DHCP on R1:** one pool per VLAN with default-router, DNS 192.168.90.10 and the first 10 addresses excluded
- **Outcome:** clients get addresses automatically, e.g. Admin-PC1 192.168.10.11 and Staff-Laptop1 192.168.70.11

Admin-PC1 receives its address by DHCP



ROUTING: OSPF

- **Point-to-point link:** R1-Core 10.0.0.1/30 and R2-Edge 10.0.0.2/30
- **OSPF** advertises all nine VLAN subnets from R1-Core to R2-Edge
- **R2-Edge** learns 192.168.10.0 to 192.168.90.0 /24 as O routes via 10.0.0.1

show ip route on R2-Edge

```
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
O    192.168.10.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.20.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.30.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.40.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.50.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.60.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.70.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.80.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
O    192.168.90.0/24 [110/2] via 10.0.0.1, 00:00:30, GigabitEthernet0/0
```

9

VLAN routes learned by R2

100%

R1–R2 ping success (5/5)

/30

10.0.0.0 link subnet

SHARED SERVICES



DNS SERVER

192.168.90.10 · DNS (UDP 53). Resolves airport.local, www.airport.local and services.airport.local



WEB SERVER

192.168.90.20 · HTTP (TCP 80). Airport portal opens at http://airport.local



FTP + EMAIL

Airport-Services 192.168.90.30 · FTP (TCP 21) and Email (SMTP/POP3, airport.local)

WIRELESS ACCESS



STAFF WI-FI

SSID Airport-Staff · VLAN 70 · WPA2-PSK (AES) · channel 6. Laptops get 192.168.70.x by DHCP



PASSENGER WI-FI

SSID Airport-Passenger · VLAN 80 · WPA2-PSK (AES) · channel 6. Gateway 192.168.80.1

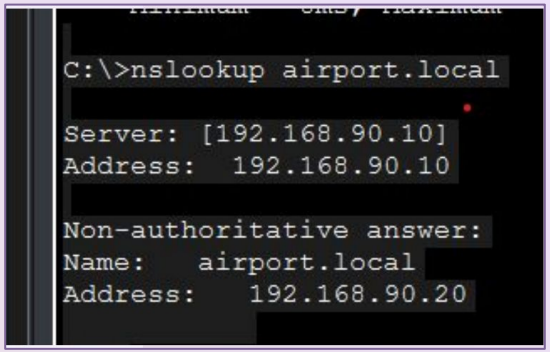
CONNECTIVITY TESTING

Source	VLAN	Gateway	DNS server .90.10	Web / Services
Admin-PC1	10	✓	✓	–
Check-in PC	20	✓	✓	–
Security-PC1	30	✓	✓	✓ ping + browser
Baggage-PC1	40	✓	✓	–
Ops-PC1	50	✓	✓	–
IT-PC1	60	✓	✓	–
Staff-Laptop1	70	✓	✓	✓ web .90.20, services .90.30
Passenger-Laptop1	80	✓	✓	✓ browser
Passenger-Laptop2	80	✓	✓	–

- **Ping results:** All recorded ping tests completed successfully (4/4 replies, 0% packet loss).
- **TTL 127** on server pings shows the packets crossed R1-Core (routed between VLANs)

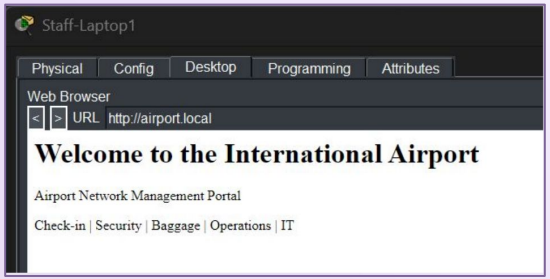
APPLICATION-LAYER SERVICES

DNS (UDP 53)



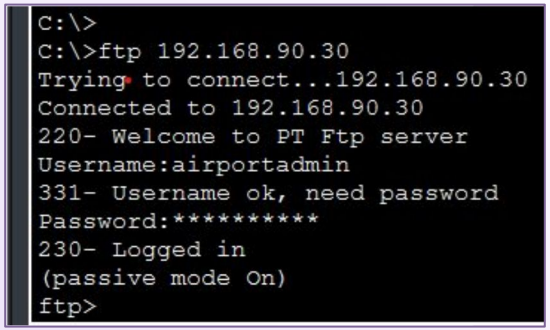
nslookup airport.local returns 192.168.90.20

HTTP (TCP 80)



Portal opens by name from staff, security and passenger devices

FTP (TCP 21)

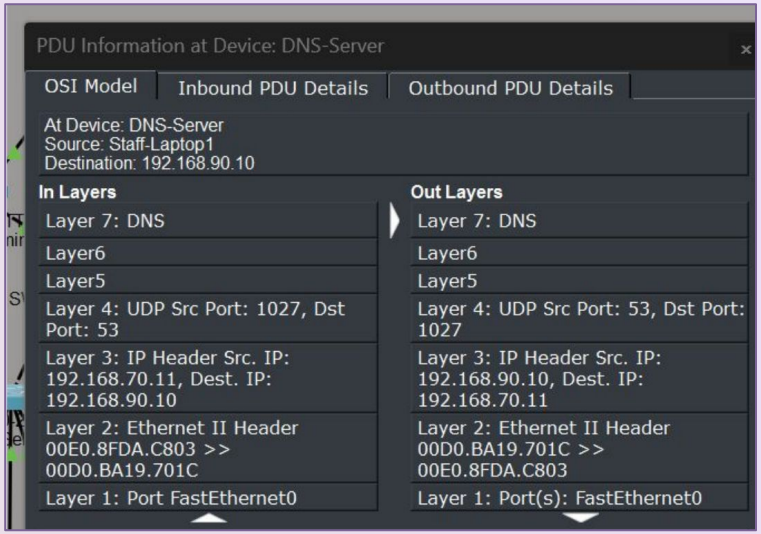


Login to 192.168.90.30 succeeds (230 Logged in)

- **Email:** SMTP and POP3 enabled on Airport-Services, domain airport.local, users admin and ops

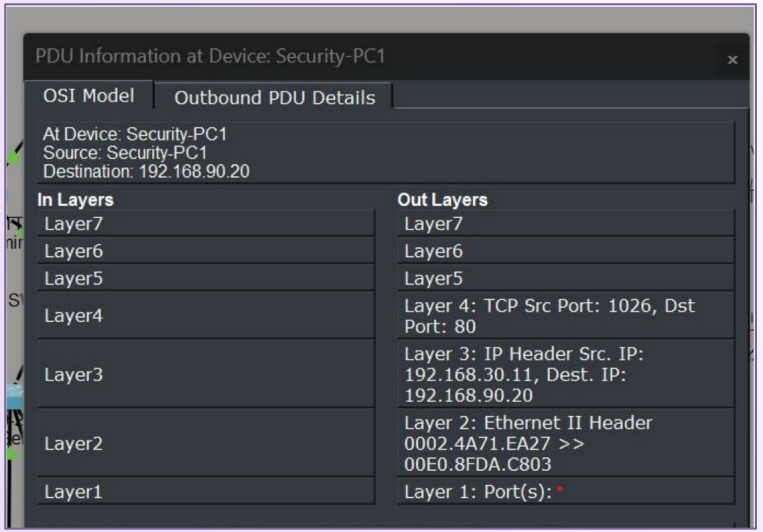
SIMULATION MODE: PACKET TRACE

DNS query (UDP): Staff-Laptop 1 to DNS-Server



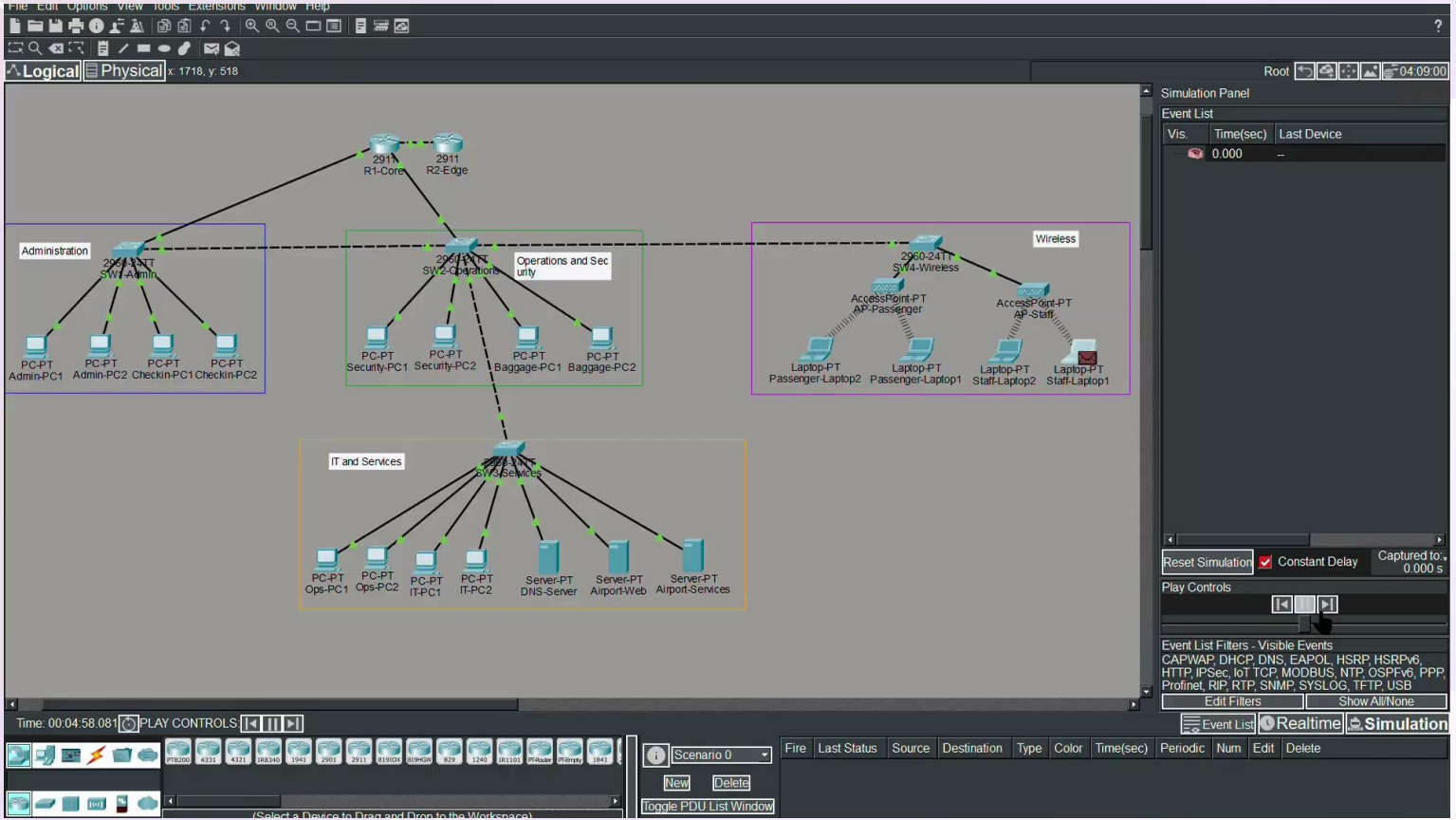
- **App/Transport:** DNS over UDP 1027 to 53
- **Network/Link:** 192.168.70.11 to 192.168.90.10, Ethernet II frame
- **Path:** AP-Staff, SW4, SW2, R1-Core, SW2, SW3, DNS-Server

HTTP (TCP): Security-PC1 to Airport-Web

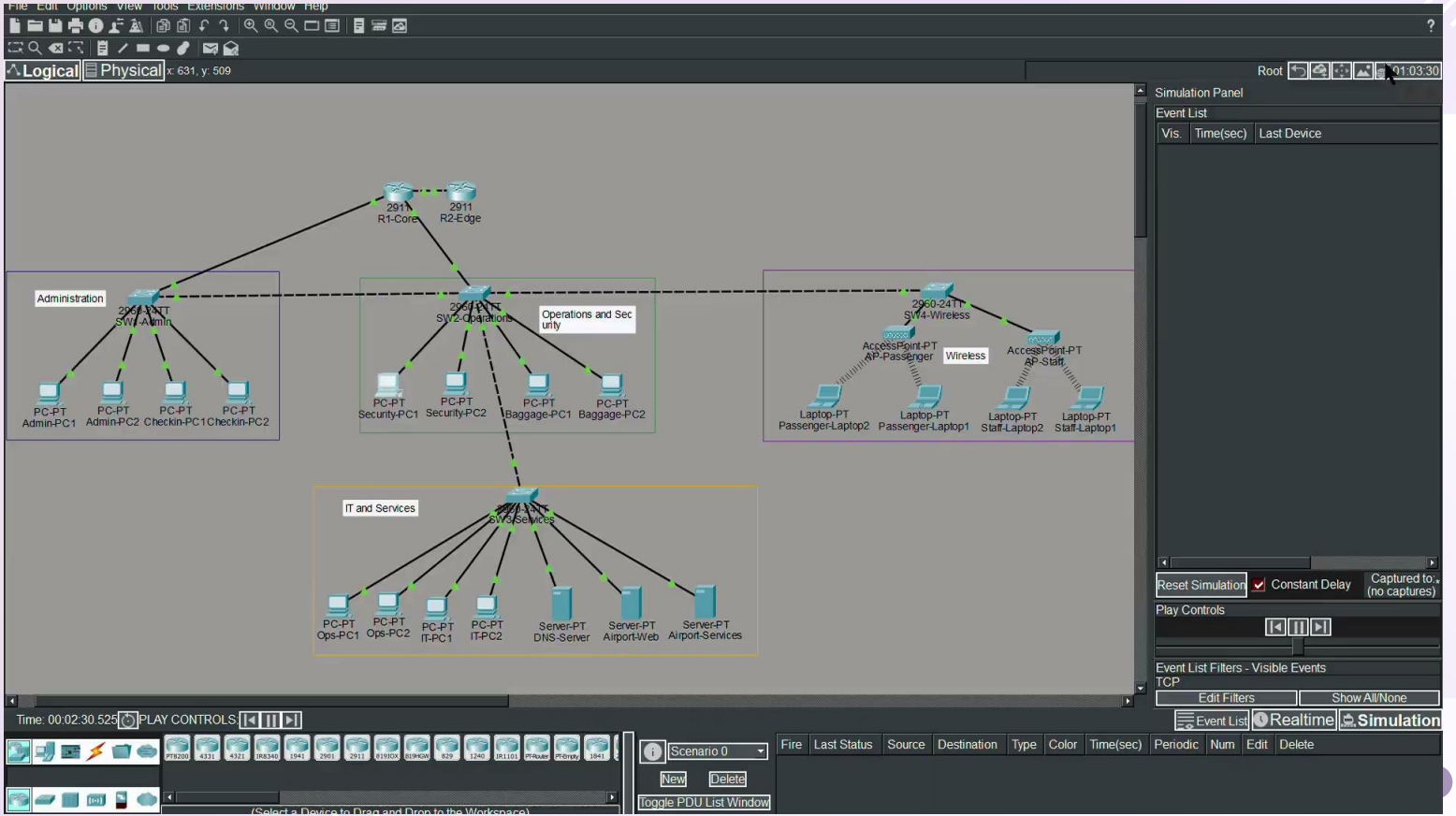


- **App/Transport:** HTTP over TCP 1026 to 80, connection opened then closed (FIN+ACK)
- **Network/Link:** 192.168.30.11 to 192.168.90.20, Ethernet II frame
- **Routing:** R1-Core forwards between VLAN 30 and VLAN 90

DNS Packet Trace Simulation



HTTP TCP Packet Trace Simulation



THANK YOU

